DEVICEOPS / PROFESSIONAL AGENTS

Autonomous Incident Triage Agent

Observe → investigate → authorize → act → verify. Built with the AWS Strands Agents SDK.

01 DEVICE & DELIVERY LAYER

Controlled LAN / local notification services

OBSERVATION SOURCES

Devices & simulators

LAN equipment: read-only probes
Standalone HTTP simulator

NODE.JS EDGE COLLECTOR

Normalize & deliver

Ping · SNMP system / ports · PJLink
Encrypted vault + durable spool
Outbound API + command journal

↑ simulator-only commands

INTEGRATION BOUNDARY

57-entry connector catalog

4 adapters implemented; 53 planned.
Physical/cloud writes disabled.

LOCAL NOTIFICATIONS

n8n → Mailpit

Email delivery + signed callback

02 DEVICEOPS APPLICATION

Next.js + TypeScript worker + PostgreSQL

telemetry

OPERATOR ENTRY POINT

Next.js operations UI + v2 API

Sessions · tenant scope · manager approvals
Collector ingestion · MCP tool endpoint

persisted state

AUTONOMOUS ORCHESTRATION

TypeScript operations worker

Poll / detect / group by explicit topology
PostgreSQL-backed pg-boss investigation jobs
Built-in simulator + durable invocation budgets

validated proposal

BACKEND OWNS AUTHORIZATION

Policy → execution → verification

Service restart: automatic when policy allows
Switch restart: manager approval + exact hash
One command; 3 fresh healthy samples or escalate

PERSISTENCE

PostgreSQL 17

Tenant RLS · evidence · runbooks · jobs · audit

investigate

03 AGENT & MODEL LAYER

Private Python service + external model API

VERIFIED LIVE MODEL PATH

Google Gemini API

gemini-3.7-flash · external to AWS
Quota-limited; no silent provider fallback

inference

PYTHON / FASTAPI

AWS Strands Agents SDK

Bounded reasoning + structured diagnosis
Isolated invocation process + usage limits
Proposes actions; cannot execute them

tool calls

STDIO MCP → SCOPED API READS

7 evidence tools

Incident · health · logs · topology
Recent changes · runbooks · capabilities
Short-lived invocation grant; no device secrets

MCP reads use tenant-scoped API

OPTIONAL / ACCOUNT-BLOCKED

Amazon Bedrock + Titan

Nova reasoning; Titan semantic runbooks

RETRIEVAL Gemini uses lexical runbook search. Optional Titan vectors are evaluated in the service; no hosted vector database is claimed.

END-TO-END INCIDENT LIFECYCLE

Observe → Detect & group → Investigate → Authorize → Execute → Verify → Report / notify

VERIFICATION SNAPSHOT · 08 SEP 2026

4 live Gemini scenarios passed on synthetic devices. Failed-recovery logic has automated coverage; its live-model run remains quota-blocked.